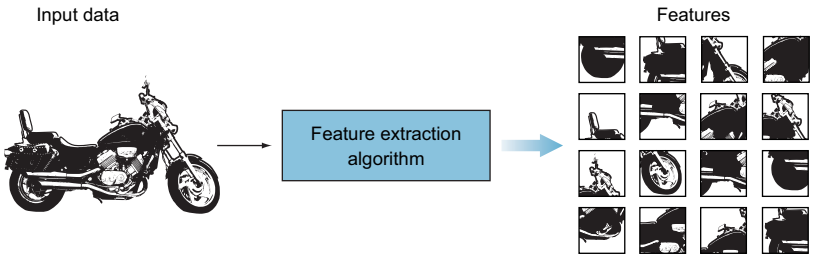
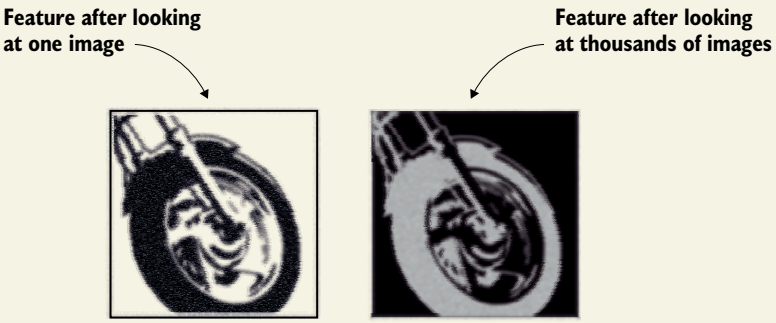


Figure 1.22 Example input image fed to a feature-extraction algorithm to find patterns within the image and create the feature vector

Feature generalizability

It is important to point out that figure 1.22 reflects features extracted from just one motorcycle. A very important characteristic of a feature is *repeatability*. The feature should be able to detect motorcycles in general, not just this specific one. So in real-world problems, a feature is not an exact copy of a piece of the input image.



Features need to detect general patterns.

If we take the *wheel* feature, for example, the feature doesn't look exactly like the wheel of one particular motorcycle. Instead, it looks like a circular shape with some patterns that identify wheels in all images in the training dataset. When the feature extractor sees thousands of images of motorcycles, it recognizes patterns that define wheels in general, regardless of where they appear in the image and what type of motorcycle they are part of.

1.6.2 What makes a good (useful) feature?

Machine learning models are only as good as the features you provide. That means coming up with good features is an important job in building ML models. But what makes a good feature? And how can you tell?